

LLM Evaluation Analytics Dashboard

Problem Statement and literature Survey

1. Problem Statement

Large Language Models (LLMs) such as GPT, Gemini, Claude, and other generative AI models are increasingly used in real-world applications. However, **evaluating and comparing their performance is challenging** because different models can produce different results depending on the prompt, task, and evaluation criteria. Existing evaluation methods often rely on manual testing, isolated metrics, or static reports, making it difficult to analyse large numbers of responses and understand the reasons behind performance differences.

Therefore, there is a need for an **interactive LLM Evaluation Analytics Dashboard** that can systematically evaluate and compare multiple LLMs using standardized prompts and datasets. The system should analyse responses based on **accuracy, relevance, coherence, factuality, safety, response time, token usage, and cost**, and present the results through meaningful visualizations.

The proposed system aims to help **developers, researchers, and organizations** understand **when, where, and how different LLMs perform**, identify strengths and weaknesses, and make informed decisions about selecting, evaluating, and improving LLM-based applications.

2. Literature Survey

1. LLM Comparator: Interactive Analysis of Side-by-Side Evaluation of Large Language Models.

Author(s) :- Minsuk Kahng, Ian Tenney, Mahima Pushkarna, Michael Xieyang Liu, James Wexler, Emily Reif, Krystal Kallarackal, Minsuk Chang, Michael Terry, and Lucas Dixon

Algorithm / Methodology :- The methodology begins with collecting test prompts and sending the same prompts to multiple LLMs. Their responses are collected and compared side-by-side using an LLM-as-a-Judge approach. The responses are evaluated based on criteria such as accuracy, relevance, coherence, helpfulness, and safety. The evaluation results are processed to generate scores and identify performance patterns. Finally, the results are presented through interactive charts, graphs, and tables to provide clear model comparisons and insights.

Advantages :- The system enables easy comparison of multiple LLMs, provides interactive and visual analysis, improves the interpretability of evaluation results, identifies performance patterns and errors, supports large-scale automated evaluation, helps developers improve models, and provides useful insights for researchers and organizations.

Disadvantages / Limitations :- The system may face limitations such as errors or bias in LLM-based evaluation, high computational and API costs, dependence on evaluation metrics and prompt design, difficulty in interpreting numerical scores, variation in model responses, challenges in managing large datasets, subjectivity in evaluating qualities such as creativity, and the continued need for human validation.

2. Political Bias in Large Language Models: A Comparative Analysis of ChatGPT-4, Perplexity, Google Gemini, and Claude

Author(s) :- TAVISHI CHOUDHARY

Algorithm / Methodology :- The study selects four LLMs—ChatGPT-4, Perplexity, Gemini, and Claude—and tests them using standardized political questions from established political quizzes. Their responses are collected, analyzed, compared for political tendencies, and presented through tables and charts. The results are used to identify possible patterns of political bias.

Advantages :- The study enables **systematic comparison**, identifies differences between LLMs, increases awareness of possible political bias, supports AI transparency, and provides a basis for future bias research and mitigation.

Disadvantages / Limitations :- The study is limited by the **number of models and questions**, prompt sensitivity, model updates, difficulty in measuring political bias, response variability, evaluation subjectivity, and limited generalization to all political topics.

3. AI Sensors and Dashboards

Author(s) :- HUBER FLORES

Algorithm / Methodology :- The paper proposes an **AI-sensor-based continuous monitoring approach** for trustworthy AI. Data is collected and prepared, the model is trained and deployed, and AI sensors continuously monitor **data quality, fairness, bias, explainability, and performance** throughout the AI pipeline. The collected information is displayed on an interactive dashboard for analysis, monitoring, and adjustment.

Advantages :- The approach promotes **trustworthy-by-design AI**, enables continuous monitoring, improves transparency, supports fairness and explainability analysis, helps detect problems early, strengthens auditing and accountability, and provides potential for automated trust management.

Disadvantages / Limitations :- The approach can be **complex and resource-intensive** because multiple sensors may be required. Continuous monitoring can increase computational overhead, while selecting suitable sampling frequencies is challenging. Sensor configuration, security risks, explainability implementation, varying system requirements, and dependence on sensor-data quality can affect reliability. Further research and standardization are also needed.

4. Screening Interface Usability with Multi-Modal LLMs: A Case Study on BOTMA Dashboards

Author(s) :- JOONWOO LEE 1 , (Member, IEEE), SEUNGHO KIM2 , (Member, IEEE), AND SCOTT UK-JIN LEE 2

Algorithm / Methodology :- The study uses a **two-stage evaluation method** in which interface screenshots are first rated by human experts and then evaluated by multiple Vision-Language Models (VLMs) using standardized prompts. A calibration method is then applied to align model scores with human ratings.

Advantages :- The method **reduces dependence on human experts**, enables fast and repeatable evaluation, supports continuous testing, and improves raw VLM scores through calibration.

Disadvantages / Limitations :- The method is limited to **static screenshots**, provides limited diagnostic feedback, may suffer from **overfitting**, focuses on a single domain, and lacks validation on broader usability benchmarks.

5. Beyond the Leaderboard: A Survey of the Science of Evaluation, Benchmarking, and Methodologies for Large Language Models

Author(s) :- SAEID SHEIKHI , LAURI LOVÉN AND PANOS KOSTAKOS

Algorithm / Methodology :- The paper uses a **survey-based methodology** that reviews and classifies existing LLM evaluation approaches, including static benchmarks, dynamic evaluation, interactive/agentive evaluation, human-in-the-loop evaluation, and LLM-as-a-Judge. It also analyzes challenges such as benchmark overfitting, data contamination, and leaderboard bias, and proposes a continuous benchmark lifecycle: **Design → Validate → Deploy → Monitor → Update → Retire**.

Advantages :- The paper provides a **comprehensive evaluation framework**, identifies limitations of traditional benchmarks, supports multiple evaluation methods, encourages continuous and real-world testing, and provides useful concepts for developing an LLM evaluation dashboard.

Disadvantages / Limitations :- The paper does not provide a **single implementable evaluation algorithm** and has limited experimental validation. Evaluation results can depend on datasets, prompts, metrics, and judge models. Human evaluation is costly, continuous evaluation requires significant resources, benchmark contamination is difficult to eliminate, and benchmark scores may not fully represent real-world LLM performance.

6. Comparative Literature Survey

Paper	Algorithm / Methodology	Advantages	Disadvantages / Limitations
LLM Comparator: Interactive Analysis of Side-by-Side Evaluation of Large Language Models.	The methodology begins with collecting test prompts and sending the same prompts to multiple LLMs. Their responses are collected and compared side-by-side using an LLM-as-a-Judge approach. The responses are evaluated based on criteria such as accuracy, relevance, coherence, helpfulness, and safety. The evaluation results are processed to generate scores and identify performance patterns. Finally, the results are presented through interactive charts, graphs, and tables to provide clear model comparisons and insights.	The system enables easy comparison of multiple LLMs, provides interactive and visual analysis, improves the interpretability of evaluation results, identifies performance patterns and errors, supports large-scale automated evaluation, helps developers improve models, and provides useful insights for researchers and organizations	The system may face limitations such as errors or bias in LLM-based evaluation, high computational and API costs, dependence on evaluation metrics and prompt design, difficulty in interpreting numerical scores, variation in model responses, challenges in managing large datasets, subjectivity in evaluating qualities such as creativity, and the continued need for human validation.
Political Bias in Large Language Models: A Comparative Analysis of	The study selects four LLMs—ChatGPT-4, Perplexity, Gemini, and	The study enables systematic comparison , identifies differences	The study is limited by the number of models and questions , prompt

ChatGPT-4, Perplexity, Google Gemini, and Claude	Claude—and tests them using standardized political questions from established political quizzes. Their responses are collected, analysed, compared for political tendencies, and presented through tables and charts. The results are used to identify possible patterns of political bias.	between LLMs, increases awareness of possible political bias, supports AI transparency, and provides a basis for future bias research and mitigation.	sensitivity, model updates, difficulty in measuring political bias, response variability, evaluation subjectivity, and limited generalization to all political topics.
AI Sensors and Dashboards	The paper proposes an AI-sensor-based continuous monitoring approach for trustworthy AI. Data is collected and prepared, the model is trained and deployed, and AI sensors continuously monitor data quality, fairness, bias, explainability, and performance throughout the AI pipeline. The collected information is displayed on an interactive dashboard for analysis, monitoring, and adjustment.	The approach promotes trustworthy-by-design AI , enables continuous monitoring, improves transparency, supports fairness and explainability analysis, helps detect problems early, strengthens auditing and accountability, and provides potential for automated trust management.	The approach can be complex and resource-intensive because multiple sensors may be required. Continuous monitoring can increase computational overhead, while selecting suitable sampling frequencies is challenging. Sensor configuration, security risks, explainability implementation, varying system requirements, and dependence on sensor-data quality can affect reliability. Further research and standardization are also needed.
Screening Interface Usability with Multi-Modal LLMs: A Case Study on BOTMA Dashboards	The study uses a two-stage evaluation method in which interface screenshots are first rated by human experts and then evaluated by multiple Vision-Language Models (VLMs) using standardized prompts. A calibration method is then applied to align model scores with human ratings.	The method reduces dependence on human experts , enables fast and repeatable evaluation, supports continuous testing, and improves raw VLM scores through calibration.	The method is limited to static screenshots , provides limited diagnostic feedback, may suffer from overfitting , focuses on a single domain, and lacks validation on broader usability benchmarks.
Beyond the Leaderboard: A Survey of the Science of Evaluation, Benchmarking, and Methodologies for Large Language Models	The paper uses a survey-based methodology that reviews and classifies existing LLM evaluation approaches, including static benchmarks, dynamic evaluation, interactive/agentive evaluation, human-in-the-loop evaluation, and LLM-as-a-Judge. It also analyzes challenges such as benchmark overfitting, data contamination, and leaderboard bias, and proposes a continuous	The paper provides a comprehensive evaluation framework , identifies limitations of traditional benchmarks, supports multiple evaluation methods, encourages continuous and real-world testing, and provides useful concepts for developing an LLM evaluation dashboard.	The paper does not provide a single implementable evaluation algorithm and has limited experimental validation. Evaluation results can depend on datasets, prompts, metrics, and judge models. Human evaluation is costly, continuous evaluation requires significant resources, benchmark contamination is difficult to eliminate, and

	benchmark lifecycle: Design → Validate → Deploy → Monitor → Update → Retire.		benchmark scores may not fully represent real- world LLM performance.
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1. Research Gap Identified

- **Lack of Unified Evaluation Platform** – Existing studies often focus on individual evaluation methods or specific metrics, but there is a need for a single platform that combines multiple evaluation dimensions in one place.
- **Limited Interactive Analysis** – Many existing evaluation approaches provide numerical scores or static reports, but offer limited interactive visualization and exploration of evaluation results.
- **Difficulty in Side-by-Side Model Comparison** – Comparing multiple LLMs across the same prompts, datasets, and evaluation criteria is not always straightforward.
- **Limited Interpretability of Evaluation Results** – Existing metrics may indicate *how well* a model performs but may not clearly explain *why* its performance differs from another model.
- **Scalability of Manual Evaluation** – Human evaluation of large numbers of LLM responses is time-consuming and difficult to perform consistently.
- **LLM-as-a-Judge Reliability** – Automated judging can improve scalability, but issues such as evaluator bias, inconsistent judgments, and sensitivity to evaluation prompts require further investigation.
- **Limited Combined Analysis of Quality, Cost and Latency** – Many evaluation studies emphasize response quality while practical deployment factors such as **API cost, token usage, and response latency** may receive less integrated analysis.
- **Insufficient Continuous Monitoring** – Evaluation is often performed as a one-time benchmark, while deployed AI applications require continuous monitoring to detect changes in performance over time.
- **Limited Generalization Analysis** – A model's performance on a particular benchmark may not represent its performance on unseen or domain-specific datasets, creating a need for broader validation.

2. References

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